Storing data in files

A computer's memory doesn't just store numbers and characters. Many more types of data can be stored, including music, pictures, and videos. But how is this data stored? And how can it be found again?

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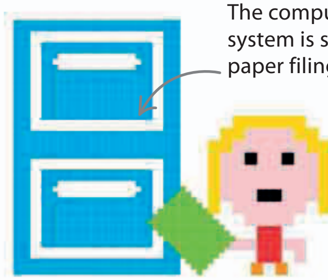
How is data stored?

When data is saved to be used later, it is put into a file. This file can be given a name that will make it easy to find again. Files can be stored on a hard-drive, memory stick, or even online—so data is safe even when a computer is switched off.



The computer's file system is similar to a paper filing system

EXPERT TIPS

File sizes

Files are essentially collections of data in the form of binary digits (bits). File sizes are measured in the following units:

Bytes (B)

1 B = 8 bits (for example, 10011001)

Kilobytes (KB)

1 KB = 1,024 B

Megabytes (MB)

1 MB = 1,024 KB = 1,048,576 B

Gigabytes (GB)

1 GB = 1,024 MB = 1,073,741,824 B

Terabytes (TB)

1 TB = 1,024 GB = 1,099,511,627,776 B

▽ **File information**

There is more to a file than just its contents. File properties tell the system everything it needs to know about a file.

Right-click on a file to see properties such as file type, location, and size

FILE PROPERTIES

The file name should be memorable	**name**	**groove**
What type of file it is, typically in three characters	**file type extension**	**mp3**
The program that can handle the file's data	**opens with**	**Music Player**
The location of the file on the computer	**full directory path**	**/Users/Jack/Music**
The file size (see the box on the left)	**size**	**50 MB**